Daniel Porter

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EDUCATION

University of California, Berkeley May 2024 – PRESENT
Bachelor of Arts in Statistics Berkeley, CA

- **Minor:** Cognitive Science & Technology
- **Coursework**: Reproducible & Collaborative Data Science, Computational Models of Cognition, R Programming, Foundations of Data Science, Experimental Design, Probability & Statistics, Introduction to Neurotechnology

San Diego Miramar College Aug 2021 – May 2024
Associate of Arts in Mathematics Studies San Diego, CA

- **Coursework**: Calculus 1/2/3, Linear Algebra, General Psychology

RESEARCH EXPERIENCE

University of California, Berkeley – Shenhav Lab Jan 2025 – PRESENT
Research Assistant Berkeley, CA

- Developed a modular inference pipeline for value-based decision-making models, integrating simulation, likelihood approximation, and parameter estimation
- Implemented a flexible Leaky Competing Accumulator (LCA) simulator from scratch with configurable collapsing bounds (linear and Weibull), variable noise, and customizable parameter regimes across experimental conditions
- Generated large synthetic datasets to train neural likelihood estimators using Mixed Neural Likelihood Estimation (MNLE) via the sbi library
- Extended the pipeline by integrating learned likelihoods with PyMC to support maximum likelihood estimation and MCMC-based inference for real participant data
- Validated the framework through parameter recovery analyses and posterior predictive checks, reproducing known behavioral patterns from prior literature on exclusive vs. inclusive choice

SKILLS

Programming : Python, C/C++, Bash, R
Systems & Tools : Linux, VMware, VirtualBox, Git, GitHub, Jupyter
Technical Areas : Statistical Modeling, Simulation-Based Inference, Computational Modeling

PROGRAMS & FELLOWSHIPS

Research Experience Pathways (Psychology) Aug 2024 – Present

- Selected for competitive funded fellowship supporting independent undergraduate research
- Awarded $6,000 towards undergraduate research project

L'SPACE NASA Proposal Writing & Evaluation Experience Jan 2024 – Apr 2024

- Collaborated on a team-based mission concept, developing a technical design and formal proposal
- Evaluated feasibility, mission constraints, and system-level considerations

NASA Community College Aerospace Scholars Jun 2023 – Aug 2023

- Completed engineering design project simulating NASA mission workflows
- Gained exposure to mission architecture and systems engineering processes